ENGINEERING GRAPHICS

I Year I Semester: MECHANICAL ENGINEERING, AERONAUTICAL ENGINEERING & EEE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6ME01	ESC	1	0	4	3	40	60	100

COURSE OUTCOMES:

At the end of the course the student should be able to:

1. Explain various commands and create drawing in AutoCAD.
2. Sketch the various curves used in engineering applications
3. Prepare orthographic projections of lines, planes by visualizing them in different positions.
4. Solve the problem of projections of solids and development of surfaces for industrial needs.
5. Construct the isometric view into orthographic views and vice versa.

UNIT- I INTRODUCTION

Introduction to Engineering Graphics: Principles and their significance,
INTRODUCTION TO COMPUTER AIDED DRAFTING: Initial Setup Commands, Draw Commands, Modify Commands, 2D Drawings-Simple Exercises.

UNIT- II ENGINEERING CURVES

Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only).
Special curves: Cycloids, Involutes.

UNIT- III ORTHOGRAPHIC PROJECTION

Principles of Orthographic Projections – conventions – first and third angle projections. Projections of points-Projection of lines inclined to both the planes. (First angle projection only)
PROJECTIONS OF PLANES: Projections of regular planes inclined to both planes.

UNIT-IV PROJECTION OF SOLIDS AND DEVELOPMENT OF SURFACES

PROJECTION OF SOLIDS: Regular Solids inclined to both planes
DEVELOPMENT OF SURFACE/SOLIDS: Theory of development, development of lateral surface with base.

UNIT- V ISOMETRIC DRAWINGS

Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections. Drawing Isometric circles, Dimensioning Isometric objects.
Conversion of Isometric view to Orthographic views and Orthographic views to isometric view.

Text Books:

1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House.
2. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication

Reference Books:

1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka “Engineering Graphics with AutoCAD” PHI publications, 2013.
2. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers.
3. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education.